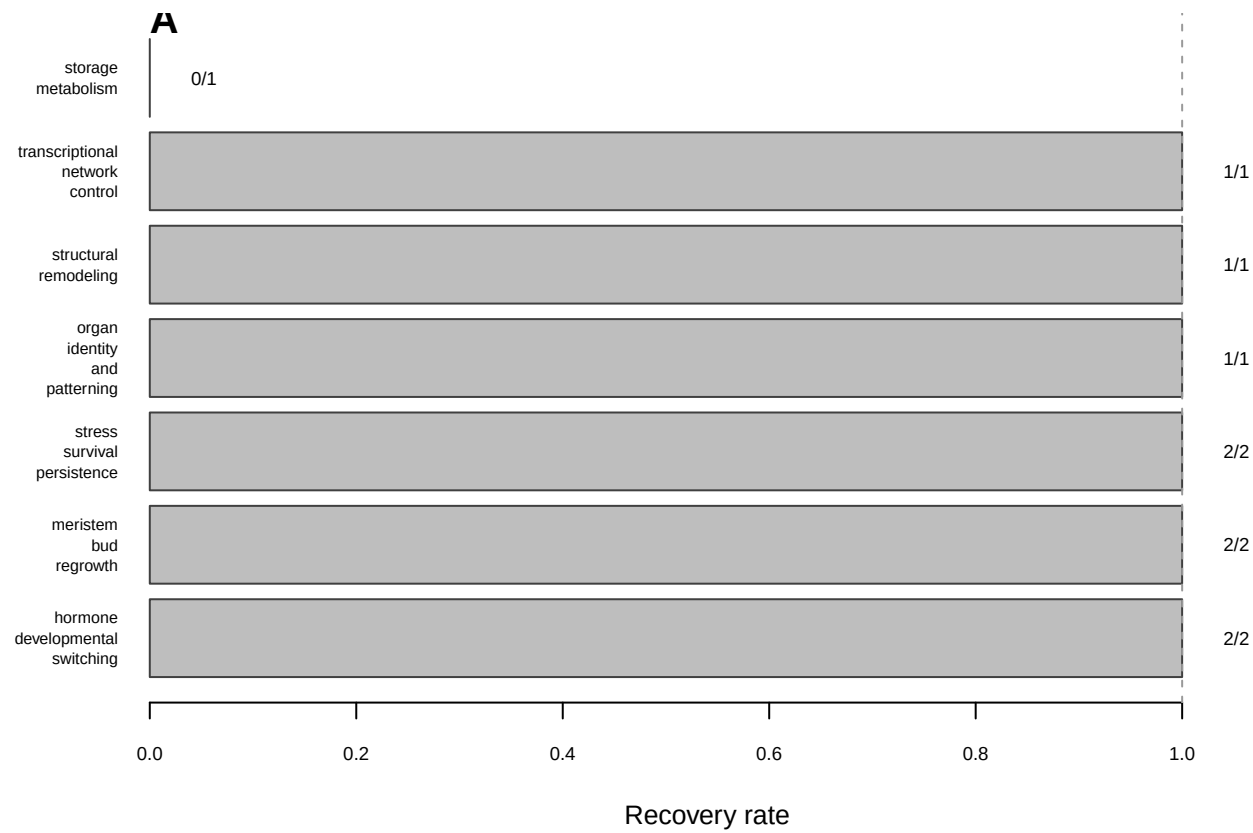
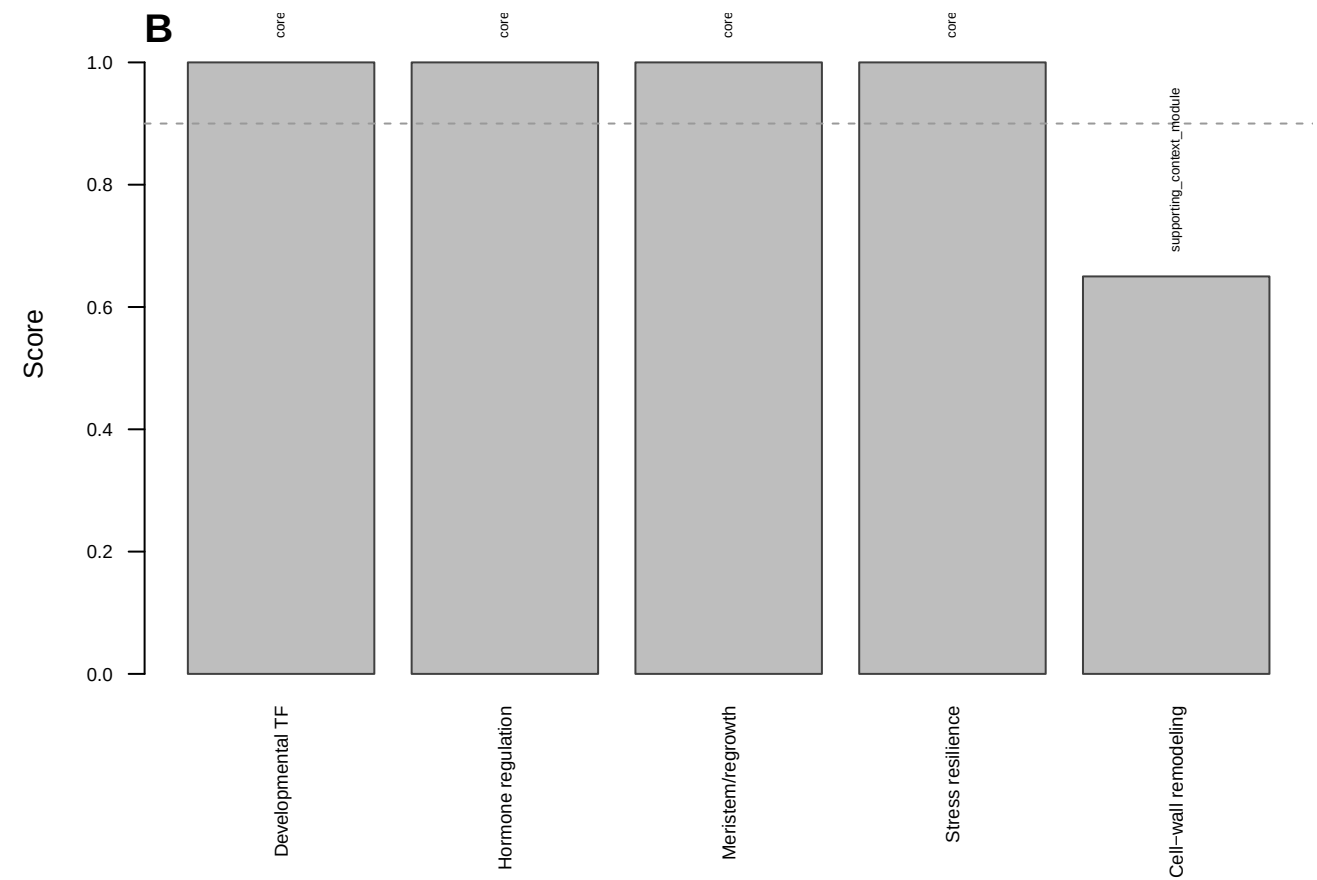


Known-category recovery



Integrated discovery score



Species support for core toolkit

core modules present across backbone species

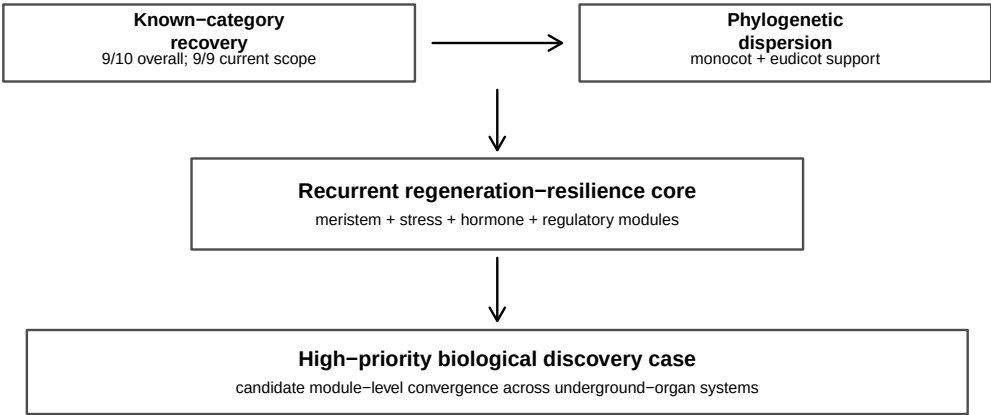
Species	Clade	Organ	Core modules	Supporting modules
Beta vulgaris	eudicot	storage_root	4	1
Brassica rapa	eudicot	root_hypocotyl_storage	4	0
Daucus carota	eudicot	storage_root	4	0
Manihot esculenta	eudicot	storage_root	4	0
Nelumbo nucifera	eudicot	rhizome	4	0
Solanum tuberosum	eudicot	tuber	4	0
Dioscorea rotundata	monocot	tuber	4	1
Musa acuminata	monocot	corm_rhizome_like_stem	4	1
Oryza longistaminata	monocot	rhizome	4	0
Panicum virgatum	monocot	rhizome	4	1
Zingiber officinale	monocot	rhizome	4	0

● present ○ not detected

D

Benchmark-supported discovery case

from recovery credibility to core toolkit hypothesis



Interpretation: supports candidate prioritization and future testing, not functional validation of individual genes.